

Section 12:

PROBLEM	POSSIBLE CAUSE	SOLUTION
BLOWER STARTS, BUT SHUTS DOWN AFTER APPROXIMATELY 10 SECONDS AND FLAME SAFEGUARD GOES INTO LOCKOUT	Non-Recycling Limit Circuit Open (Between terminals 3 & P for Fireye E110 or terminals 6 & 7 for Honeywell RM7800)	Combustion Air Switch Open
		Auxiliary Low Water Cutoff Open. Manual reset may be required (May not be present on all boilers).
		High Limit Pressure or Temperature Switch Open. Manual reset may be required.
		High Gas Pressure Switch Open. Manual reset may be required (Gas Burners Only)
		Low Gas Pressure Switch Open. Manual reset may be required. (Gas Burners Only)
		Low Oil Pressure Switch Open. (Oil Burners Only)
		Low Oil Temperature Switch Open. (Heavy Oil Burners Only)
		High Oil Temperature Switch Open. (Heavy Oil Burners Only)
		Low Atomizing Air Pressure Switch Open (Oil Burners Only)
		Blower Motor Contactor Auxiliary Contact Open
		Any other Controls in the Non-Recycling Limit Circuit may be open. Check the Wiring Diagram for controls not mentioned above.
BLOWER STARTS BUT AFTER 10 MINUTES OF PURGE, FLAME SAFEGUARD GOES INTO LOCKOUT	High Fire Purge Limit not Made. (Between Terminals D & 8 for Fireye E110 or 5 & 19 for Honeywell RM7800)	If Firing Rate Motor does not Drive to High Fire Position: a. Check 24V Power Supply From Transformer b. Check for Binding Linkage c. Check for Faulty Firing Rate Motor
		If Firing Rate Motor does Drive to High Fire: a. Check for Faulty or Improperly Adjusted High Fire End Switch inside Firing Rate Motor. b. Check if the Purge Air Pressure Switch is Open (if supplied)
Note: Run/Check switch on flame safeguard must be in the run position for the startup sequence to proceed.		

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HIGH FIRE PURGE IS COMPLETED AND FIRING RATE MOTOR RETURNS TO LOW FIRE, WAITS 10 MINUTES, AND FLAME SAFEGUARD GOES INTO LOCKOUT	Low Fire Start Limit not Made (Between Terminals M & D for Fireye E110 or 5 & 18 for Honeywell RM7800)	Check for Faulty or Improperly Adjusted Low Fire End Switch inside Firing Rate Motor
Note: Run/Check switch on flame safeguard must be in the run position for the startup sequence to proceed.		
PURGE IS COMPLETED AND IGNITION ON LIGHT TURNS ON BUT NO PILOT FLAME IS ESTABLISHED	Faulty Flame Safeguard	Check for 115V between safeguard ignition terminal and neutral (5 & L2 for Fireye E110 firing light oil or gas, 6 & L2 for Fireye E110 firing heavy oil, 8 & L2 for Honeywell RM7800 firing light oil or gas, L1 & L2 for Honeywell RM7800 firing heavy oil)
	Improper Gas Supply To Pilot Regulator	a. Upstream valves must be open b. Excessive pressure to regulator can cause regulator lock-up.
	Improper Electrode Spark Gap	Adjust electrode spark gap
	Broken Electrode Insulator	Replace electrode insulator
	Blower Dampers are not Closed	Close blower dampers
	Faulty Pilot Solenoid Valves	Replace faulty pilot solenoid valves
	Faulty Ignition Transformer	Replace ignition transformer
	Faulty or Disconnected High Voltage Lead from Ignition Transformer to Electrode	Connect lead from ignition transformer to electrode with electrode.

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PILOT FLAME IS ESTABLISHED AND PROVEN FOR 10 SECONDS, BUT MAIN FLAME NOT ESTABLISHED, FUEL VALVE LIGHT ON	Run/Check Switch in "Check" Position	Set to "Run" Position
	Faulty Flame Safeguard	Check for 115V between flame safeguard main fuel terminal and neutral (7 & L2 for Fireye E110 and 9 & L2 for Honeywell RM7800)
	Fuel Control Valve Actuator Defective	Replace Fuel Control Valve Actuator
	Auxiliary Main Fuel Valve Actuator Defective	Replace Auxiliary Main Fuel Valve Actuator
	Manual Oil Valve at Burner Closed (Oil Only)	Manual Oil Valve at Burner must be Open
	Incorrect Burner Adjustment (Oil)	(See Adjustments Section) a. Oil Pump Pressure b. Oil Preheat (Heavy Oil Only) c. Modulating fuel valve d. Oil Nozzle Carboned up or plugged – Check and clean e. Atomizing Air not Sufficient to support Low Fire f. Cold oil in Line – Heavy Oil Only
	Normally Open Vent Valve not Closed (Gas Only)	a. Coil Burned out or Defective b. Mechanical Part of Valve Inoperable
	Checking Cock Closed (Gas)	Open Valve
PILOT FLAME ESTABLISHED AND PROVEN; MAIN FLAME ATTEMPTS TO LIGHT BUT PROGRAMMER CONTROL GOES INTO ALARM CONDITION	Incorrect Burner Adjustment (Gas)	a. Check for proper incoming gas supply to main regulator – Check Fire Test Sheet b. Check regulated gas pressure – Check Fire Test Sheet c. Check for slipped or disconnected linkage d. Make sure gas control valve is set properly for low fire lightoff.
	Momentary Opening of Non-Recycling Interlock Circuit (Between Terminals 3 & P for Fireye E110 and 6 & 7 for Honeywell RM7800)	a. Check Combustion Air Switch – Should be Closed when Blower Starts b. Check Low Oil Pressure Switch when Oil Valves Open – Check Oil Pressure Gauge for Momentary Drop of Pressure c. Check Low Gas Pressure Switch when Gas Valves Open – Check Regulated Gas Pressure Gauge for Momentary Drop of Pressure

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MAIN FLAME NOT STABLE AFTER IGNITION ENDS	Incorrect Air-Oil Ratio (Oil Firing)	(See Adjustments Section for Adjustment Procedure) a. Check for slipped or disconnected linkage. b. Make sure Oil Control Valve is set properly. c. Check for Proper Oil Pressure to Oil Nozzle d. Check for Belt Slippage on Compressor or Oil Pump e. Check for Proper Oil Temperature to Nozzle (Heavy Oil) f. Check for Suction Leak on Suction Line Creating Air Pockets in Oil Supply g. Check for Water in Oil Supply – Suction Line should be 6" or more off Bottom of Tank h. Check for Dirty Oil Strainer
	Incorrect Air-Gas Ratio (Gas Firing)	(See Adjustments Section for Adjustment Procedure) a. Check for slipped or disconnected linkage b. Make sure gas control valve is set properly c. Check for dirt or pipe scale in Main Regulator – Make sure that a dirt leg is in ahead of regulator or a strainer d. Check for ruptured diaphragm in regulator e. Check for a plugged vent line on regulator f. Make sure regulator external sensing line (if applicable) is installed correctly. g. Make sure normally open vent valve between safety shutoff gas valves closes tight when safety shutoff gas valves are energized
SAFETY SHUTOFF VALVE(S) DO NOT SHUT OFF WHEN OPERATING LIMITS OPEN	Safety Shutoff Valve(s) Mechanically Stuck Open	Check for obstructions and manually close valve.

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MAIN FLAME LIGHTS, IGNITION IS OFF, BUT FIRING RATE MOTOR DOES NOT RESPOND TO MODULATING CONTROLLER.	Low Fire Hold Control Open – Located In Boiler Shell Sensing Water Temperature	a. Check for proper setting (180-190°F) before switch closes b. Make sure switch closes when temperature rises
	Note: Firing switch on panel set for “Manual” and manual potentiometer on panel set “Open” will override low fire hold control and allow firing rate to go to high fire.	
	Modulating Controller Calling for High Fire but Firing Rate Doesn’t Respond	a. Check controller for proper setting of desired steam pressure/water temperature to be maintained b. Check potentiometer coil and wiper arm
	Note: Manual Potentiometer must be set to “Open” position on panel and Firing switch on panel set to “Automatic” for modulating controller to operate.	
	“Manual-Automatic” Switch on Panel Inoperable	Replace switch
	Manual Potentiometer on Panel Inoperable	Replace Potentiometer
FLAME SIGNAL DROPS BELOW MINIMUM WHILE MAIN BURNER IS FIRING AND MODULATING FROM LOW TO HIGH FIRE AND VICE VERSA. FLAME SAFEGUARD GOES INTO LOCKOUT.	Burner not Adjusted Properly with Flame Pattern Changing Too Abruptly or Varying Too Much for Reliable Monitoring by Scanner	Re-Adjust Flame (See Adjustment Section)
	Scanner Cell Weak	Replace Scanner Cell
	Flame Safeguard Defective	Replace Flame Safeguard
MAIN FLAME SHUTS OFF DURING NORMAL FIRING CYCLE WITH FLAME SAFEGUARD GOING INTO LOCKOUT	Opening of Non-Recycling Interlock Circuit (Between terminals 3 & P on Fireye E110 or 6 & 7 on Honeywell RM7800).	Check combustion air switch and low fuel pressure switches for abnormal conditions.
	Main Oil/Gas Flame Blows Out While Firing	Adjust fuel air ratio. (See adjustments section)